

Advanced Packaging, From First Principles

A ground-up rebuild of what the SemiAnalysis four-part series tried to teach.

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1. What this document is

Advanced packaging is the least-well-explained important topic in semiconductors. Every few months a new brand name — CoWoS-L, FOEB, Foveros Omni, X-Cube — appears, gets dropped into articles that assume you already know what bumps and TSVs and interposers are, and the topic becomes a blur of acronyms.

It shouldn't be a blur. The entire field answers one question:

How do you get data on and off a silicon die fast enough, and with low enough energy per bit, without making the chip impossibly expensive?

Every packaging technology — old, new, and unreleased — is a different way to trade cost against *interconnect density*. Once you hold that axis in your head, all the brand names collapse into points on a single ladder.

This document builds that ladder from scratch. It then walks every significant commercial technology — TSMC's InFO and CoWoS and SoIC families, Intel's EMIB and Foveros line,

Samsung's FOSiP and I-Cube and X-Cube, ASE's FoCoS and FOEB, Amkor's SLIM, Sony's image sensors, YMTC's NAND — and assigns each one a rung. At the end you will see why the industry is now blurring the lines between 2.5D, 2.3D, 2.1D, and 3D, and why that blur is a victory, not a problem.

No prior knowledge is assumed beyond "a chip is a piece of silicon with transistors on it."

2. The two crises that forced packaging to matter

Advanced packaging is recent — as a commercial priority, really the last five years. For most of the transistor era nobody cared how chips were packaged, because chips themselves were improving so fast that the packaging overhead looked negligible. Two things changed that simultaneously.

2.1 The transistor-cost crisis

Moore's Law, as originally stated, has two limbs. Transistor *density* doubles roughly every two years, and — this is the economic limb — transistor *cost* halves at the same rate. For decades both held together. The density limb still mostly works: a 5nm transistor is smaller than a 7nm transistor. The cost limb broke around the introduction of FinFETs.

Relevant numbers:

- A TSMC N7 wafer costs roughly \$9,500.
- A TSMC N5 wafer costs roughly \$16,000.
- Transistors per mm² improved maybe $1.8\times$ between those two nodes.
- Cost per transistor *got worse* at 5nm.

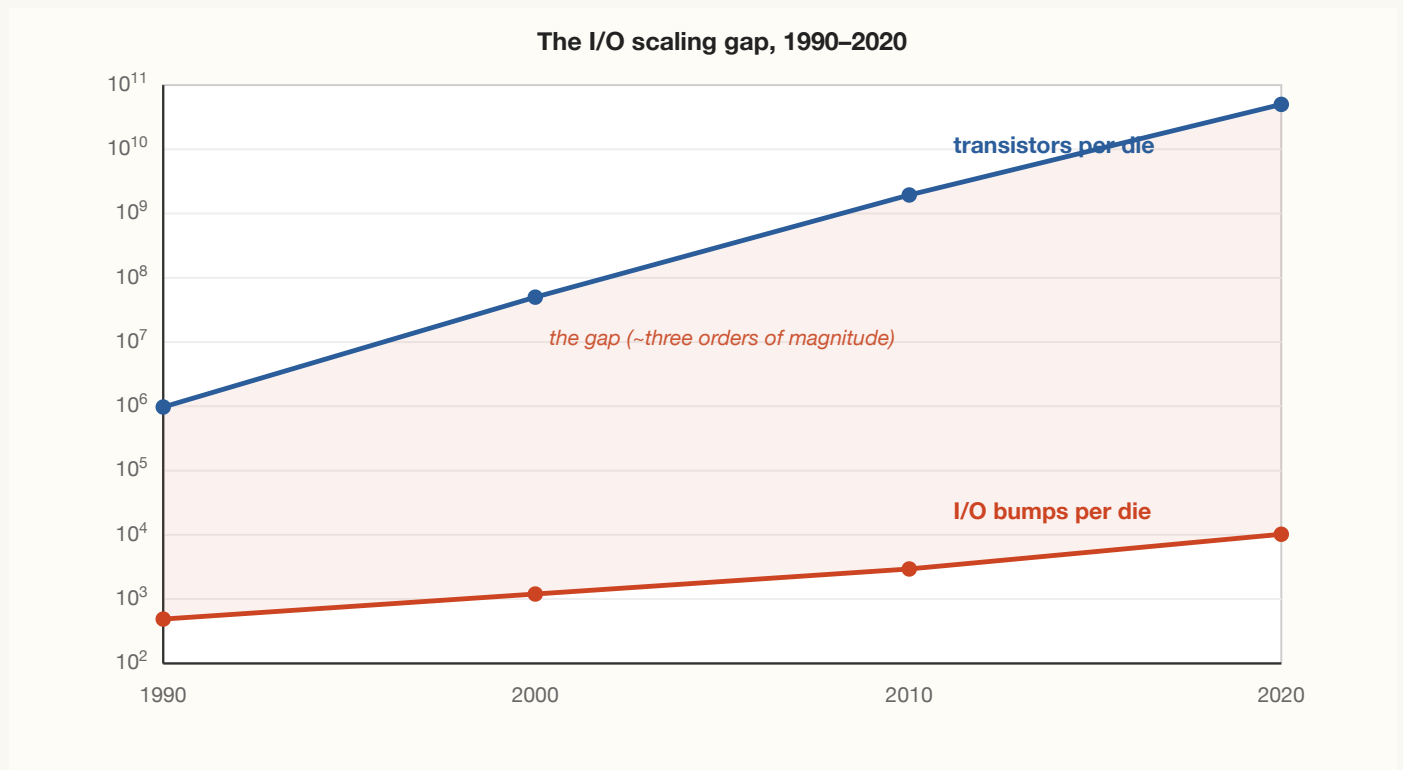
Why? Three compounding reasons. EUV lithography is expensive. A modern node uses $\sim 35\%$ more process steps than the previous one. And yield suffers as dies get larger, because defect density per mm² is roughly constant — a single defect kills an expensive chunk of silicon.

The implication is simple and nasty: **you can no longer get cheaper by moving to a newer node**. You get more performance per mm², but each mm² is more expensive. The old strategy of "wait two years, port to the next node, watch costs fall" stopped working at FinFET.

2.2 The I/O crisis — "pad-limited" designs

A chip is not just transistors. It is also *pads* — the electrical contacts where signals and power enter and leave the die. Pads sit at the surface of the chip. Each pad takes a certain minimum area, set by the *bump pitch*, the center-to-center spacing of the solder contacts that sit on each pad.

Here's the historically painful number. Flip-chip bump pitch went from $\sim 200\ \mu\text{m}$ in the 1990s to $\sim 130\ \mu\text{m}$ by the 2010s. That's about a $2\times$ improvement in bump density. In the same window, transistor density improved by roughly $2,000\times$.



Transistor count per die has grown roughly three orders of magnitude faster than the number of I/O bumps on the same die. The shaded region is the problem advanced packaging exists to solve.

The practical effect is that a modern chip's floor plan often looks like a small logic region surrounded by a large ring of bump pads. The logic could shrink to the next node, but the pad ring *can't*, because the package can't handle any finer a pitch. So the die stays artificially large, and the expensive new node doesn't buy you a smaller die — it buys you extra silicon you didn't want. This is the condition called **pad-limited**.

Pad-limiting is the fingerprint of the whole advanced-packaging problem. Everything the industry has invented — fan-out, interposers, bridges, TSVs, hybrid bonding — is some way of breaking the pad-pitch floor.

2.3 The reticle wall

There is a third ceiling, separate from transistor costs and bump pitch, which is harder than either. ArF excimer lithography uses a reticle of fixed maximum size: $33\text{ mm} \times 26\text{ mm}$, or **858 mm²**. That is the biggest area you can expose in one shot. You can draw a bigger layout, but you cannot pattern a die larger than that in a single exposure.

Nvidia's datacenter GPUs have been bumping against the reticle limit for roughly five years. The H100 is $\sim 814\text{ mm}^2$. The A100 was $\sim 826\text{ mm}^2$. You literally cannot buy a bigger monolithic logic die from TSMC.

Even if you could, yield punishes you. Defect density is roughly constant per mm² of wafer — so for a fixed process, defects per die scale linearly with die area. An 800 mm² die gets, in round numbers, eight times as many defects per die as a 100 mm² die. Good dies per wafer fall faster than area. The arithmetic gets vicious fast; the point is that large dies are a yield penalty on top of a cost penalty.

2.4 The workload shift

Two crises walked in from the transistor side. A third, bigger one walked in from the workload side: chips stopped being just CPUs.

An Apple A15 or M2 is, by area, mostly not CPU. The CPU cores are a small fraction. The rest is cache SRAM, image signal processor, video codec blocks, neural engine, graphics, memory controllers, and display pipes. The workloads Apple ships that silicon for — camera, ML inference, media, screen — do not benefit much from general-purpose CPU cores.

Scale this up. A Google TPU is mostly a big systolic array of multipliers plus a wall of SRAM. An Amazon Trainium is a similar story. A Tesla Dojo training tile is almost entirely custom compute plus on-die memory plus an enormous bandwidth fabric. A Nvidia H100 is logic plus 80 GB of HBM3 memory sitting in the same package. AMD's Infinity Cache trick saved the Navi 21 GPU from a 384-bit memory bus by putting 128 MB of on-die SRAM between the compute and the GDDR.

The generalisation: modern workloads — AI, imaging, networking, recommendation — want a *heterogeneous* mix of silicon types. Logic at the most aggressive node. SRAM at an older, cheaper node (SRAM stopped scaling at N5). DRAM in its own special process. Maybe photonics or analog RF on top. These processes are mutually incompatible. You cannot fabricate them on the same die.

So you need a way to have them all live together in the same package.

2.5 The chiplet resolution

Put the three crises together and you get a forcing function.

- Transistors don't get cheaper anymore. You cannot afford to put things that don't need the latest node on the latest node.
- Dies can't get much bigger. The next generation of compute has to come from more silicon per package, not a bigger single die.
- Workloads want a mix of process types. The single-die answer doesn't even meet the requirement.

The answer is **chiplets**: take what used to be a monolithic die, break it into smaller pieces built on different processes, and re-assemble them inside a single package.

The benefits are straight arithmetic. Small dies yield better. Older processes for I/O and SRAM cost less. Designs can be reused across SKUs — AMD famously covered three market segments with three chiplet types, where Intel needed five monolithic designs. You can put *more compute than fits in a reticle* into one package.

The cost — and this is the whole subject of this document — is that you now have to connect the chiplets. Connections between dies have historically been 100–1,000× worse than connections within a die along three axes: *density* (wires per mm), *energy per bit*, and *latency*. Every time a signal hops off a die and onto a package and back onto another die, you pay enormous tax.

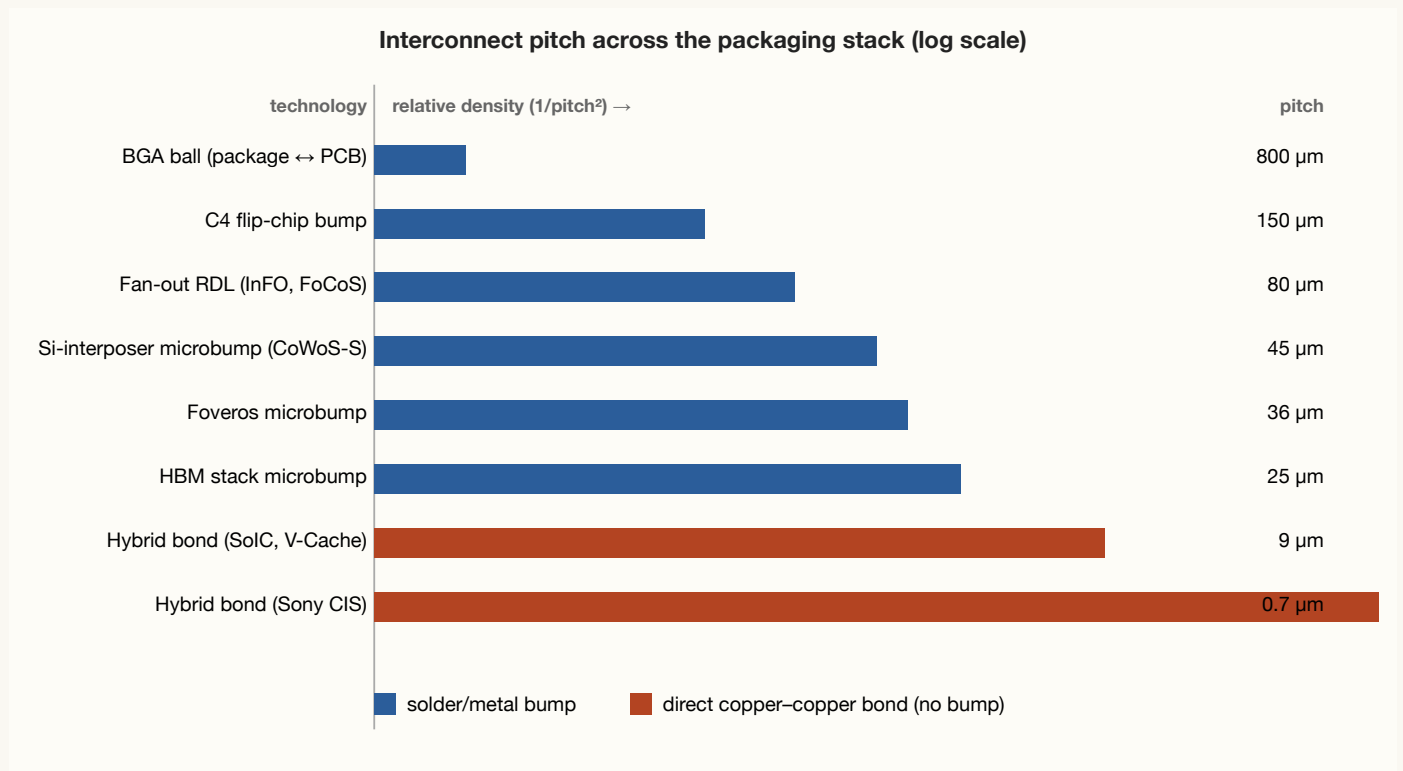
Advanced packaging is the project of making between-die connections approach within-die connections. That is the entire job. Everything in this document is about the methods, and about where each method sits on the single axis that matters: interconnect density.

3. The single axis that organizes everything

3.1 Bump pitch is the master variable

Hold one number in your head: **pitch**. It is the center-to-center distance between adjacent electrical contacts at some interface — between die and substrate, die and interposer, die and die. Measured in micrometres (μm , "microns").

Density scales as $1/\text{pitch}^2$. Halving pitch gives four times as many connections in the same area. Everything in advanced packaging is, in the end, about driving pitch down.



Each row down the table is roughly $2\times$ denser than the one above. From BGA ball (package to motherboard) to Sony's hybrid-bonded image sensor is about six orders of magnitude of density.

Every advanced packaging technology is a way of achieving some row on this chart at some cost. The lower you want to go, the more exotic (and expensive) the process. "CoWoS" and "EMIB" and "SoIC" are just brand names for specific ways of reaching specific rows.

3.2 The dimensionality axis (and its marketing inflation)

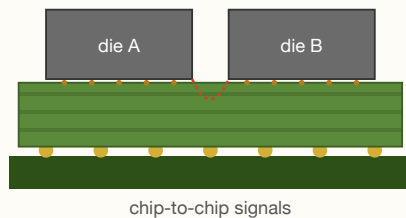
Running *perpendicular* to the pitch axis is a second axis: how the chips are physically arranged relative to each other. This is what "2D / 2.5D / 3D" refers to. It's a separate dimension and gets conflated with density constantly.

- **2D:** chips lie flat next to each other on an organic substrate. The wires between them run through the substrate.
- **2.5D:** chips lie flat next to each other on top of a passive silicon slab (an *interposer*). The wires between them run through the interposer, which — because it's silicon — can carry far denser wiring than organic substrate. The interposer then connects downward to the organic substrate through another set of bumps.

- **3D:** chips are *stacked* vertically. Active silicon sits on active silicon. Signals travel up and down through the silicon (through-silicon vias, "TSVs", or direct bonds) rather than sideways.

2D vs 2.5D vs 3D — the dimensionality axis

2D



organic substrate 2.5D silicon interposer (passive) die A die B chip-to-chip signals through *silicon interposer* 3D base die (die B) top die (die A) signals go up/down chip-to-chip signals through *the silicon itself*

"2.5D" literally means: there is a passive silicon layer between the dies and the organic substrate. It is not an intermediate density level; it is a separate architectural choice (flat vs. stacked vs. interposer-mediated flat).

ASIDE ON 2.1D / 2.3D

When you read about "2.1D" vs "2.3D" in a trade article, silently translate to: *organic substrate with extra lithographic routing*, or *organic substrate with a silicon bridge embedded*. The "point-something-D" terminology tells you nothing about the actual structure and nothing at all about density, which is what matters. Always go back to pitch.

3.3 The ladder of interconnect

Now the payoff. Every significant commercial packaging technology sits on one of seven rungs. Here is the whole ladder, coarsest to finest. The rest of this document elaborates each one.

Rung 0 — Wire bond. Gold or copper wires run from pads on the die edge up to lead frames on the package. 50–100 μm pitch along the die edge only. Used on memory, power management, analog, cheap microcontrollers — anywhere pitch doesn't matter. Not covered further.

Rung 1 — Flip-chip on organic substrate. The die is flipped active-side down; C4 solder bumps join die pads to an organic laminate substrate; underfill is injected to lock them in place. Pitch: 100–200 μm . The substrate is copper core plus Ajinomoto build-up film ("ABF"), the

material used in every modern desktop CPU substrate. This is the baseline. "A packaged chip" means this unless someone specifies otherwise.

Rung 2 — Fan-out (FOWLP). Eliminate the substrate. Embed the bare die(s) in an epoxy mold compound, then pattern *redistribution layers* (RDL) — thin-film copper wires on polymer — directly on top of the mold, fanning out to a wider array of external bumps. Pitch: 60–90 μm . L/S (line/space of the copper wires): 2–15 μm . Cheaper than Rung 1 for small packages because there's no separate substrate to buy. TSMC calls its fan-out **InFO**; ASE calls theirs **FoCoS**; Amkor calls theirs **WLFO**; Samsung calls theirs **FOSiP**. Apple A-series chips live here.

Rung 3 — Silicon bridge in substrate. An organic substrate (or fan-out) with a cavity, into which you embed a tiny piece of patterned silicon. Two dies above the cavity talk to each other through this silicon bridge, which gives silicon-class density *only where needed* (at the die-to-die interface). Everywhere else the package stays cheap organic. Intel's **EMIB** is this. ASE's **FOEB** is a fan-out variant (glass carrier). TSMC's **InFO-LSI** is a fan-out variant too. Silicon bridge = cheaper than a full silicon interposer; denser than plain fan-out; no reticle limit because you can scatter many bridges across a big package.

Rung 4 — Silicon interposer. A large slab of passive silicon (usually a few thousand mm^2 , stitched across multiple reticles) sits under *all* the dies. The silicon has fine copper wiring patterned on its top surface for die-to-die connections, and TSVs through its body to pass signals and power down to an organic substrate below. Pitch to the dies above: 40–55 μm . Used by every high-volume AI GPU with HBM: Nvidia A100/H100, Google TPU v4/v5, Amazon Trainium, Intel Gaudi. TSMC calls it **CoWoS-S**. Samsung calls it **I-Cube**.

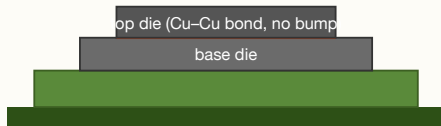
Rung 5 — 3D active stacking with TSVs and microbumps. Go vertical. Thin a die to $\sim 30 \mu\text{m}$ or less; drill TSVs through it; put microbumps on both faces; bond it face-down to another die. Signals now travel straight up through the stack. This is how **HBM** is built — a base logic die with 8 or 12 DRAM dies stacked on it. It is also how Intel **Foveros** works — a base die with logic dies stacked on top. Pitch: 25–50 μm . Because the stack is vertical, you can't easily use solder reflow — you need **thermocompression bonding** (TCB, see §5.2) to place dies one at a time under pressure and heat.

Rung 6 — 3D hybrid bonding. Eliminate the bumps entirely. Polish the two mating silicon surfaces to near-atomic flatness. Bring them together in a clean environment. The copper pads on each surface fuse directly to the copper pads on the other through diffusion at modest temperature. No solder. No underfill. No gap. Pitch: sub-10 μm for logic, 0.7 μm demonstrated in Sony's CMOS image sensors. TSMC brands it **SoIC** (the CPU-to-SRAM bond in AMD's 3D V-Cache is this). Intel brands it **Foveros Direct**. Samsung brands it **X-Cube**. YMTC uses it wafer-to-wafer for NAND (**XStacking**). Sony's stacked image sensors were first to market.

The ladder of interconnect — seven rungs, coarsest (bottom) to finest (top)

RUNG 6

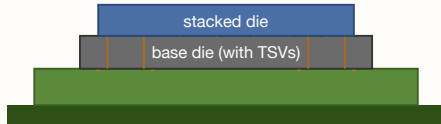
Hybrid bond
<10 μm pitch
SoIC, X-Cube



TSMC SoIC (AMD 3D V-Cache, MI300 tiles)
Intel Foveros Direct
Samsung X-Cube; YMTC XStacking
Sony CIS: 0.7 μm pitch demonstrated

RUNG 5

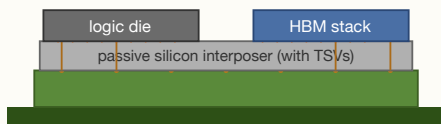
3D TSV stack
25–50 μm pitch
HBM, Foveros



HBM2/HBM3 stacks (Samsung, SK Hynix, Micron)
Intel Foveros (Lakefield, Meteor Lake)
Intel Foveros Omni (copper pillar variant)
Must be bonded by TCB, not reflow

RUNG 4

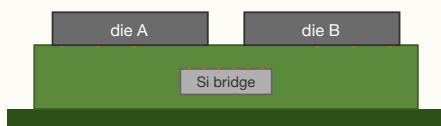
Silicon interposer
40–55 μm pitch
CoWoS-S, I-Cube



TSMC CoWoS-S (Nvidia A100/H100, TPU v4)
Samsung I-Cube (Baidu AI)
Amazon Trainium, Intel Gaudi, AMD Vega
Reticle-stitched to ~3 $\times$ reticle area

RUNG 3

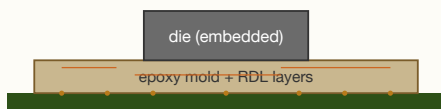
Silicon bridge
40–55 μm pitch
EMIB, FOEB, LSI



Intel EMIB (Sapphire Rapids, Ponte Vecchio)
ASE FOEB (AMD MI250X)
TSMC InFO-LSI (Apple M1 Ultra UltraFusion)
Silicon density only where needed; cheaper than full interposer

RUNG 2

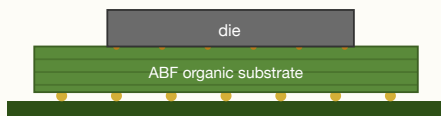
Fan-out (FOWLP)
60–90 μm pitch
InFO, FoCoS, WLFO



TSMC InFO (Apple A-series, M-series)
ASE FoCoS, Amkor WLFO, Samsung FOSiP
TSMC InFO-SOW (Tesla Dojo; wafer-scale)
No substrate — RDL patterned on the mold itself

RUNG 1

Flip-chip on ABF
100–200 μm pitch
baseline



Every desktop/server CPU & GPU substrate
Intel Core, AMD Ryzen, Nvidia GeForce GPUs
AMD EPYC, Intel Xeon (multi-chiplet via substrate only)
The "normal" package you get by default

RUNG 0

Wire bond
50–100 μm pitch
die edge only



Memory, MCUs, analog, power ICs
Not an advanced packaging technology;
included for completeness

The seven-rung ladder. Each rung up gives roughly $2\times$ more connections per mm^2 at $2\text{--}10\times$ more cost. Almost every current chip is one of these or a combination of two (e.g. AMD MI300 is Rung 4 + Rung 6).

Each rung up the ladder costs more (often multiplicatively) and delivers more bandwidth per mm^2 of interface. The industry's job — and the last decade of advanced packaging innovation — has been to push the cost curve leftward so rungs that used to be unaffordable become affordable for merchant products.

Before we zoom into the individual rungs, two more bodies of material: the common layers that show up across all of them (ABF, RDL, TSVs) and the bonding-method ladder that cuts across the interconnect ladder.

4. Substrates, RDL, TSVs — the workhorse layers

Every package has a stacked hierarchy of layers. Top to bottom:

- **The die** — the piece of silicon with transistors.
- **Near-die interconnect** — microbumps, copper pillars, or direct bond. (Rungs 5–6 stop here; they bond die to die.)
- **Interposer or bridge**, if any — Rungs 3–4 insert a piece of silicon here.
- **Fan-out RDL**, if any — Rung 2 inserts thin-film copper wiring here.
- **Organic substrate** — a laminate of copper-clad core + ABF, 4–20 layers thick, present in every non-wafer-scale package.
- **BGA balls** — big solder balls that connect the finished package to a PCB ($\sim 800\ \mu\text{m}$ pitch).
- **The PCB** — the system board.

4.1 ABF organic substrates

ABF — Ajinomoto build-up film — deserves a paragraph. It's a thin polymer sheet produced by the Japanese company Ajinomoto (yes, the monosodium glutamate company; the film was an industrial byproduct). Every modern flip-chip substrate laminates alternating layers of ABF and copper around a central glass-fibre core. Line/space achievable on ABF is about $10\ \mu\text{m}$ today; experimental coreless variants get to 3–6 μm (Unimicron has demonstrated 3 μm ; Cisco's research with 10 routing layers at 6 μm is published).

ABF supply has been a real constraint. Ajinomoto is the dominant producer. When AI-GPU packaging demand exploded in 2023, ABF allocation became a back-channel determinant of accelerator supply — which is a reminder that a 30 μm polymer film sitting under a \$30,000 GPU is load-bearing infrastructure.

4.2 RDL (redistribution layer)

RDL is shorthand for "thin-film copper wires patterned lithographically on a polymer dielectric." Three or four layers are typical in a standard fan-out package. TSMC's InFO-SoIS (used for high-end networking silicon) goes up to 14 RDL layers.

Key parameter: **line/space (L/S)**. This is the width of the copper wire and the gap between adjacent wires. "2 μm L/S" means 2 μm wires with 2 μm gaps — a density of ~ 250 wires per mm.

The approximate L/S frontier today:

Substrate type	L/S	Examples
Standard build-up ABF	10 μm	Desktop CPUs

Substrate type	L/S	Examples
Coreless ABF (research)	6 μm	Cisco; Unimicron at 3 μm
Core fanout	15 μm	Budget fan-out
Advanced fanout	2 μm	AMD RDNA 3 GPU, MediaTek networking
EMIB bridge (1st gen)	5 μm	Kaby Lake G (2018)
EMIB bridge (future)	2 μm	Roadmap
Amkor SLIM	0.4 μm	First layer only; flagship density
ASE SPIL NTI	0.5 μm	First layer only; flagship density

That last cluster — SLIM, NTI, next-gen EMIB — is where the high-density edge of organic/fan-out sits right now. It is asymptotically approaching silicon-interposer density for the fraction of the package that cares. That's why \$7 will argue the line between "organic" and "silicon" packaging is dissolving.

4.3 TSVs (through-silicon vias)

A TSV is a copper cylinder drilled vertically through a silicon wafer. It turns silicon into a substrate — signals can pass through it rather than just across it. TSVs are how:

- A silicon interposer (Rung 4) connects to the organic substrate below.
- DRAM dies in an HBM stack (Rung 5) connect to each other vertically.
- Intel Foveros passes signals from the top logic die down through the base die to the substrate.

Typical TSV diameter: 5–10 μm . Pitch: 17–40 μm . Building them requires the same fab equipment as logic itself (deep-reactive-ion etch, Cu electroplating, CMP). That matters commercially: **only a fab can build a TSV**. OSATs (Outsourced Semiconductor Assembly and Test houses — ASE, Amkor, SPIL) can assemble around existing TSV'd silicon, but they can't make the silicon itself.

This is why TSMC's packaging business is a fab business. It is also why Intel's foundry packaging arm is a competitive lever Intel likes — Intel can make TSVs in-house, and TCB tools, which means they control the whole Foveros stack without handing off.

4.4 Chip-first vs. RDL-first (a process distinction that matters)

For fan-out and interposer processes there's a build-order question with real economic consequences.

Chip-first. Place the bare dies on a carrier, overmold them in epoxy, then pattern RDL over the top. Simpler. Cheaper. But if the RDL/substrate has a defect, you've already committed good dies to it — yield loss is "good die + bad substrate = waste."

RDL-first (chip-last). Build the RDL or substrate first, test it, then bond the dies. You get a known-good substrate before you commit dies. Harder to execute (thin-film structures must be handled through test), more expensive, but avoids the yield trap.

TSMC's InFO is chip-first. CoWoS is RDL-first. Most AI-class products need RDL-first because the substrates are big and complex enough that yield losses on them are non-negligible, and the dies going on top cost thousands of dollars each. For mobile applications (Apple A-series), chip-first InFO is fine because the packages are smaller and the dies are cheaper.

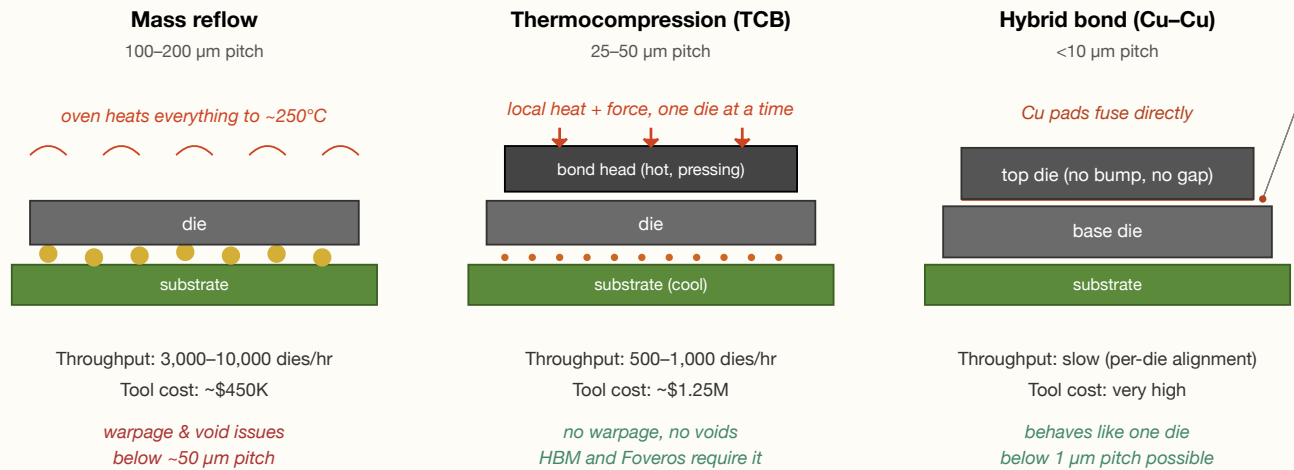
4.5 Reticle stitching

An interposer made from a single reticle exposure is capped at 858 mm². Modern AI packages need much more — H100's full interposer is ~1,200 mm²; Blackwell's is roughly 3× reticle. The trick is **reticle stitching**: expose the interposer with multiple reticle shots, aligned to stitch seamlessly so that wiring can cross reticle boundaries without breaking. TSMC pioneered 3×-reticle interposers for CoWoS; it is non-trivial and remains a technical moat.

5. The bonding-method ladder

Cross-cutting the interconnect-density ladder is a second technical axis: *how you actually join two surfaces together*. There are three generations; each rung on the interconnect ladder above tends to force a specific generation.

Three bonding generations



Three generations of joining silicon to silicon, in order of increasing density. Each one lets you fall off the previous one's density cliff. TCB is the enabling technology for HBM stacks and Intel's Foveros line; hybrid bonding is what made AMD's 3D V-Cache and Sony's stacked image sensors possible.

5.1 Mass reflow (solder)

The classical method. Place many dies on a substrate robotically (3,000–10,000 dies per hour per tool). Run the assembly through a reflow oven — a conveyor heating everything to $\sim 250^\circ\text{C}$, melting the solder, letting it bond to the pads, then cooling. Inject underfill, cure. Fast, cheap, high throughput. Works up to the 100- μm class of pitches. Tool cost $\sim \$450\text{K}$.

The problem is that at microbump pitches ($< 50 \mu\text{m}$), two things break.

- **Warpage.** Different materials have different coefficients of thermal expansion (CTE). Heating the whole assembly to reflow temperature warps the die and substrate differently; when the solder freezes the joints are uneven. Some bumps end up contacting; others don't.
- **Voids.** Surface oxidation on the solder can leave tiny bubbles in the joint. At 100 μm pitch a small void in one joint is inconsequential. At 25 μm pitch it's a dead connection.

Above some density, mass reflow stops working. AMD's Fiji GPU (2015) and some early interposer products got burned by exactly these failure modes.

5.2 Thermocompression bonding (TCB)

The next rung. Instead of heating everything in an oven, TCB places *one die at a time* with a tool head that applies:

- Local heat (just the die being placed, not the whole substrate).
- Mechanical pressure (forces the bumps to deform into good contact).

- Sometimes ultrasonic vibration (breaks the oxidation layer on the solder so clean metal meets clean metal).

The die is held in place while the joint forms, then the head lifts. Cycle time: a few seconds per die. Throughput: 500–1,000 dies per hour — 3–10× slower than reflow. Tool cost: ~\$1.25M.

Why bother? Because warpage goes away. The substrate stays cool; only the local joint sees thermal stress. Bonds are void-free and gap-variation-free. 25 μm pitch becomes reliable.

Who uses TCB. Intel is the heaviest adopter — ~300 tools today, doubling with the new Malaysia facility. Intel's whole architectural strategy (EMIB, Foveros, Foveros Omni) depends on TCB reliability; they've been co-developing TCB with vendors for a decade. TSMC uses TCB *only* for HBM stacking, not for their logic products; logic packaging is InFO-based and stays with mass reflow where possible. All HBM vendors (Samsung, SK Hynix, Micron) use TCB for HBM3's 12-die stacks with 30 μm -thin dies — you simply can't reflow-bond a stack like that without it buckling.

Tool vendors: **BE Semiconductor (Besi)**, **ASM Pacific Technology**, **Kulicke & Soffa**. Each has a slightly different niche — ASMP dominant in mobile PoP, K&S strong in HBM-style placements, Besi picking up the high-end Intel-style mixed-pitch work. TCB tool orders are a leading indicator of advanced packaging volume.

5.3 Hybrid bonding (direct Cu–Cu)

Skip solder entirely. The two mating surfaces are finished with a dielectric (usually SiO_2 ;) in which copper pads are embedded flush with the surface. Both surfaces are polished to near-atomic flatness — chemical-mechanical polish to sub-nm RMS roughness. They are cleaned, activated (a plasma treatment), brought into contact. The dielectrics fuse first through Van der Waals forces; a mild anneal (~200–400°C) then causes the copper pads to diffuse into each other and form a continuous copper interconnect across the interface.

No solder. No gap. The two dies behave as a single piece of silicon for signalling purposes. Pitches below 10 μm for logic. Sony has demonstrated 0.7 μm for CIS.

Hybrid bonding is what makes Rung 6 possible. It is the *only* way to drive pitch below the microbump floor. The catch: it is extremely unforgiving of contamination (a single particle ruins the bond) and needs near-fab-level cleanliness in the packaging step. That's part of why the boundary between fab and pack is blurring — see §7.

5.4 Which bonding method at which rung

Interconnect rung	Typical pitch	Bonding method
1 — Flip-chip on ABF	150 μm	Mass reflow
2 — Fan-out	60–90 μm	Mass reflow
3 — Silicon bridge	40–55 μm	Mass reflow or TCB
4 — Silicon interposer	40–55 μm	Mass reflow (dies to interposer); TCB (HBM on top)
5 — 3D TSV stack	25–50 μm	TCB
6 — 3D hybrid bond	<10 μm	Hybrid bond (direct Cu–Cu)

Memorise this table. When someone says "AMD's V-Cache uses SoIC" you now know, without anyone saying it, that the technology is hybrid bonding. When someone says "HBM3 stacks are 12-high," you know that TCB is on the BOM.

6. The vendor map

This is where SemiAnalysis's four-part series drowns its readers. Instead of presenting brand names cold, here they are, each tagged with its rung.

6.1 TSMC

TSMC's naming is logical once you see the structure: InFO is the fan-out family (Rung 2); CoWoS is the interposer family (Rung 4); SoIC is the hybrid-bonding family (Rung 6).

InFO family (Rung 2, fan-out, chip-first):

- **InFO-R** — the baseline. Apple A-series and M-series chips. "InFO with RDL."
- **InFO-PoP** — package-on-package. DRAM stacked on top of the InFO logic package. Apple's approach before M-series.
- **InFO-oS** (on-substrate) — fanout placed on top of an organic substrate, for larger or higher-I/O configurations.
- **InFO-SoIS** (System on Integrated Substrate) — heavy fanout with up to 14 RDL layers. Networking silicon that wants substrate-level routing complexity.
- **InFO-SOW** (System on Wafer) — wafer-scale fanout. Tesla Dojo D1 tile is reportedly this.
- **InFO-LSI** (Local Silicon Interconnect) — fanout with silicon bridges embedded for die-to-die links. Apple M1 Ultra's "UltraFusion" interconnect is reportedly this — a Rung 2+3

hybrid.

CoWoS family (Rung 4, silicon interposer):

- **CoWoS-S** — the original and still the highest-volume. Passive silicon interposer, RDL patterned lithographically. Every major AI GPU with HBM.
- **CoWoS-R** — RDL-first variant with an organic-based interposer component. Cheaper but less dense.
- **CoWoS-L** — newer variant using a mix of silicon bridges and fanout-style RDL. Nvidia Blackwell uses this. It is functionally a Rung 3+4 hybrid.

SoIC family (Rung 6, hybrid bonding). Two variants by flow:

- **CoW** (chip-on-wafer) — mount singulated dies onto a wafer, bond. AMD 3D V-Cache is CoW SoIC at 17 μm pitch.
- **WoW** (wafer-on-wafer) — bond two full wafers, singulate afterwards. Graphcore's IPU-BOW.

6.2 Intel

Intel's naming tracks architectural intent: EMIB is silicon bridge (Rung 3), Foveros is TSV stacking (Rung 5), Foveros Direct is hybrid bonding (Rung 6), and Co-EMIB combines the 3D and bridge techniques.

- **EMIB** (Embedded Multi-die Interconnect Bridge) — Rung 3. A silicon bridge embedded in an ABF substrate cavity. First deployed 2018 on Kaby Lake G; Sapphire Rapids uses it extensively. Bridge L/S has gone from 5 μm $\rightarrow$ 2 μm across generations; pitch from 55 μm $\rightarrow$ 40 μm . Intel claims near-100% yield on these bridges, challenging ASE's 80–90% marketing for the comparable FOEB.
- **Foveros** — Rung 5, 3D TSV stacking, active-on-active. First deployed 2020 on Lakefield (50 μm pitch). Ponte Vecchio uses it with 47 chiplets combined via EMIB (36 μm pitch). Meteor Lake uses a later Foveros generation for its compute/GPU/IO tiles.
- **Foveros Omni** (a.k.a. **ODI**, Omni-Directional Interconnect) — Rung 5, more flexible variant. Uses copper pillars in addition to or instead of microbumps, allowing the top die to have a larger footprint than the base die. 36 μm pitch; Meteor Lake uses this.
- **Foveros Direct** — Rung 6. Intel's brand for hybrid bonding, positioned for future products.
- **Co-EMIB** — composite. Multiple Foveros 3D stacks stitched together with EMIB. A Rung 3+5 hybrid.

6.3 Samsung

- **FOSiP** (Fan-Out System in Package) — Rung 2. Tesla HW4.0 uses this.
- **I-Cube** — Rung 4, silicon interposer. Competes with CoWoS-S. Baidu's Kunlun AI accelerators.
- **X-Cube** — Rung 6, hybrid bonding. Positioned against TSMC SoIC.

6.4 ASE / SPIL

- **FoCoS** (Fan-Out Chip-on-Substrate) — Rung 2.
- **FOEB** (Fan-Out Embedded Bridge) — Rung 3. Glass-carrier variant of silicon-bridge-in-fanout. AMD MI200/MI250X uses this.
- **SPIL NTI** — high-end Rung 2, 0.5 μm L/S on the first RDL layer.

6.5 Amkor

- **WLFO** (Wafer-Level Fan-Out) — Rung 2.
- **SLIM** — high-end Rung 2, 0.4 μm L/S on the first RDL layer.

6.6 Sony and YMTC (the non-logic pioneers)

Worth a separate mention because they were there first.

Sony has been shipping hybrid-bonded CMOS image sensors since 2016. Two-stack (pixel die over circuitry die), three-stack (pixel $\rightarrow$ DRAM buffer $\rightarrow$ circuitry), and forthcoming four-stack (separating the pixel transistor from the photodiode). Pitches reach 0.7 μm — finer than anything in the logic world. CIS is where the industry learned that hybrid bonding works at scale.

YMTC uses hybrid bonding wafer-to-wafer for NAND, a process they call **XStacking**. The idea: fabricate the NAND array and the CMOS periphery (the control logic) on separate wafers optimised for each, then bond them. This gave YMTC a real density advantage over Samsung, Micron, SK Hynix, and Kioxia for a couple of generations; competitor architectures are only now catching up.

6.7 One-page summary table

Brand	Vendor	Rung	Substrate	Bond	Example product
InFO-R	TSMC	2	Mold + RDL	Reflow	Apple A15/A16
InFO-oS	TSMC	2	Mold + RDL on ABF	Reflow	Networking silicon
InFO-SOW	TSMC	2 (wafer-scale)	Wafer-level mold	Reflow	Tesla Dojo
InFO-LSI	TSMC	2 + 3	Mold + Si bridge	Reflow	Apple M1 Ultra
CoWoS-S	TSMC	4	Si interposer	Reflow (+ TCB for HBM)	Nvidia A100/H100, TPU v4/v5
CoWoS-R	TSMC	3–4	RDL interposer	Reflow	Mid-range AI
CoWoS-L	TSMC	3 + 4	Si bridges + RDL	Reflow + TCB	Nvidia Blackwell
SoIC CoW	TSMC	6	—	Hybrid bond	AMD V-Cache, MI300
SoIC WoW	TSMC	6	—	Hybrid bond (wafer-to-wafer)	Graphcore IPU-BOW
EMIB	Intel	3	ABF + Si bridge	TCB	Sapphire Rapids, Ponte Vecchio
Foveros	Intel	5	TSV stack	TCB	Lakefield, Meteor Lake
Foveros Omni	Intel	5 (mixed)	TSV + Cu pillar	TCB	Meteor Lake
Foveros Direct	Intel	6	—	Hybrid bond	Future
Co-EMIB	Intel	3 + 5	ABF + Si bridge + TSV stack	TCB	Future server SoCs
FOSiP	Samsung	2	Mold + RDL	Reflow	Tesla HW4.0
I-Cube	Samsung	4	Si interposer	Reflow (+ TCB)	Baidu Kunlun
X-Cube	Samsung	6	—	Hybrid bond	Future
FoCoS	ASE	2	Mold + RDL	Reflow	Various
FOEB	ASE	3	Fanout + Si bridge (glass)	Reflow	AMD MI200/MI250X
SPIL NTI	ASE/SPIL	2 (fine)	Mold + 0.5 μ m RDL	Reflow	High-end fanout

Brand	Vendor	Rung	Substrate	Bond	Example product
WLFO	Amkor	2	Mold + RDL	Reflow	Various
SLIM	Amkor	2 (fine)	Mold + 0.4 μm RDL	Reflow	High-end fanout
XStacking	YMTC	6	—	Hybrid bond (W2W)	YMTC 3D NAND
CIS hybrid	Sony	6	—	Hybrid bond	Stacked image sensors

7. Why the lines are blurring — and it's a good thing

SemiAnalysis's fourth article argues that the conventional taxonomy (2D / 2.1D / 2.3D / 2.5D / 3D) is dissolving. They're right, but they explain it as categories getting muddled. The better framing is: **the category axis was always wrong**, and what's actually happening is that several technologies are converging on the same rung of the pitch ladder from different starting points.

Concretely:

- Fan-out with embedded silicon bridges (InFO-LSI, FOEB) reaches Rung 3 from the fan-out direction.
- EMIB reaches Rung 3 from the organic-substrate direction.
- Coreless ABF substrates with 3–6 μm L/S (Unimicron, Cisco research) reach *toward* Rung 3 from the commodity-organic direction.
- CoWoS-L combines silicon bridges with organic RDL to reach Rung 3–4 from the silicon-interposer direction.

Five different names. One functional sweet spot: "*somewhere between organic-only and silicon-interposer, dense enough for die-to-die logic links, cheap enough not to need full interposer.*" The "2.1D / 2.3D" vocabulary tries to differentiate these by how they got there. The rung they got to is the same.

A modern AI package is often *hybrid* in exactly this way. AMD's MI300 is a working example: SoIC hybrid-bonded tiles (Rung 6) on top of logic+IO dies (Rung 5 TSVs), sitting next to HBM stacks on a silicon interposer (Rung 4), all on an ABF organic substrate (Rung 1) on a PCB. Five rungs in one device. There is no "one technology."

The industry has stopped trying to win with a single flagship packaging approach and is now assembling packages from whichever rung is cheapest or densest for each sub-job. This is, notably, how PCBs have worked for 40 years. Packaging is finally approaching the modularity of PCB assembly — which is the appropriate destination for a field whose job is to be a better, denser PCB.

THE HIDDEN THESIS

The real story of the 2020s in advanced packaging is not that any one technology won. It is that the *cost curve for every rung has shifted downward*, so that multi-rung packages have become economic. A 2015 AI accelerator could use a silicon interposer only if it was priced at \$10,000+ (Nvidia P100 / Google TPU). A 2024 AI accelerator can afford a silicon interposer *plus* hybrid-bonded SRAM *plus* TSV-stacked HBM in the same device. That's not a taxonomy shift; it's a cost-curve shift.

8. The product atlas

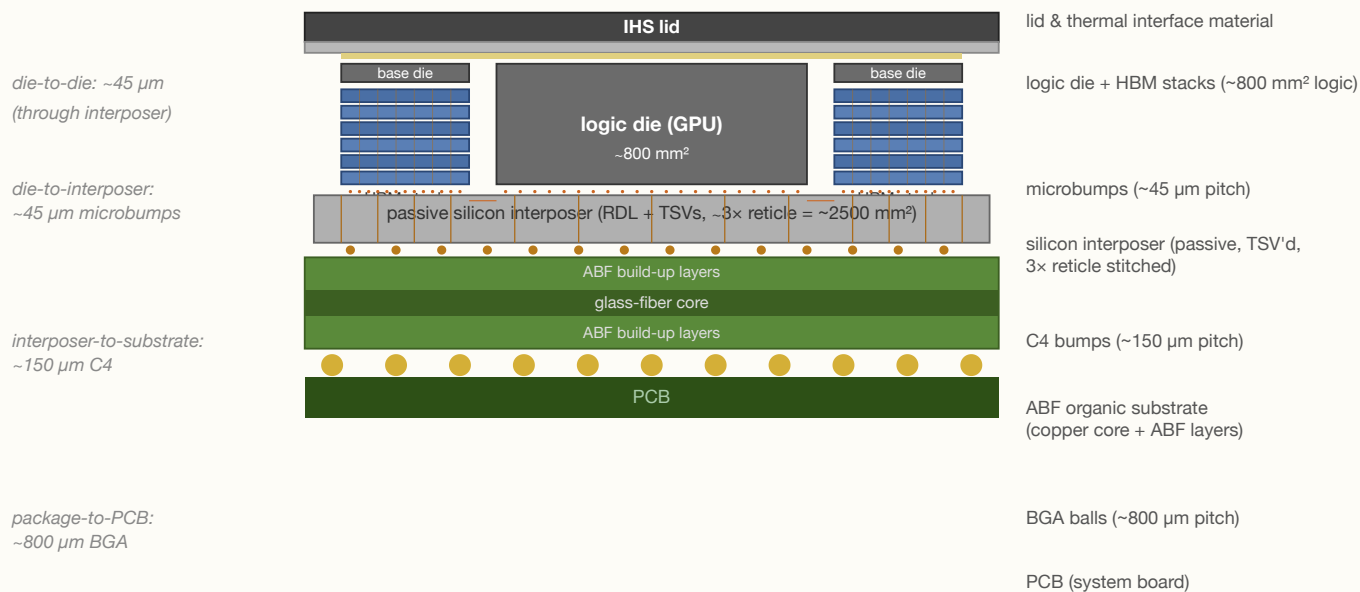
Where the famous products actually sit on the ladder:

- **Nvidia A100** — CoWoS-S. $\sim 826 \text{ mm}^2$ logic die, $6 \times$ HBM2e, reticle-stitched interposer.
- **Nvidia H100** — CoWoS-S with larger reticle-stitched interposer. $\sim 814 \text{ mm}^2$ logic die, $5 \times$ HBM3.
- **Nvidia Blackwell (B100/B200)** — CoWoS-L. Two logic dies joined by silicon bridges, $8 \times$ HBM3e, RDL fan-out elsewhere.
- **AMD MI250X** — ASE FOEB. Two GPU dies on a fan-out-with-silicon-bridge substrate, plus HBM.
- **AMD MI300** — TSMC CoWoS-S base, with SoIC hybrid-bonded logic tiles on top of IO-die. A Rung 4 + Rung 6 sandwich.
- **AMD 3D V-Cache (Zen 3+)** — TSMC SoIC CoW. 64 MB of L3 SRAM bonded on top of a CCD at $17 \mu\text{m}$ pitch.
- **Apple A15/A16/A17** — TSMC InFO-R. Single SoC in a fanout package.
- **Apple M1 Ultra / M2 Ultra** — TSMC InFO with a silicon bridge (Apple-branded "UltraFusion") joining two die halves. InFO-LSI.
- **Tesla Dojo D1 training tile** — TSMC InFO-SOW, wafer-scale fanout. 25 D1 dies on one reconstituted wafer.
- **Tesla HW4.0 (FSD inference)** — Samsung FOSiP.
- **Google TPU v4** — TSMC CoWoS-S.

- **Amazon Trainium / Graviton3** — CoWoS-S.
- **Intel Kaby Lake G** (2018) — first shipping EMIB product.
- **Intel Lakefield** — first Foveros, 50 μm pitch.
- **Intel Ponte Vecchio** — 47 chiplets, EMIB + Foveros (36 μm pitch).
- **Intel Sapphire Rapids** — 4 tiles joined by EMIB at 55 μm pitch, alongside 100 μm standard flip-chip for the rest of the connections. A mixed-pitch package.
- **Intel Meteor Lake / Arrow Lake** — Foveros Omni with mixed-pitch interfaces (130, 100, 36 μm).
- **HBM3 modules (Samsung/SK Hynix/Micron)** — Rung 5 TCB stacking, 12 dies of 30 μm thickness.
- **Sony stacked CMOS image sensors** — Rung 6 hybrid bond, 0.7 μm pitch.
- **YMTC NAND** — Rung 6 XStacking wafer-to-wafer bond.

The canonical AI-GPU package — the thing that every H100 or B200 or TPU owes its existence to — looks like this when you pull it apart layer by layer:

Anatomy of a CoWoS-S package with HBM — the canonical AI accelerator



Every H100 and every TPU looks like this. From bottom up: PCB, BGA balls, ABF organic substrate, C4 bumps, silicon interposer (reticle-stitched, with TSVs), microbumps, logic die flanked by HBM stacks, TIM, lid. The interposer is what makes this Rung 4. The HBM stack itself is Rung 5. Put differently: a single package contains three different tiers of interconnect density operating in parallel.

Once you can read that atlas — "CoWoS-L with HBM3e and NVLink-C2C" decodes to "Rung 3+4 with silicon bridges plus organic RDL, carrying an HBM Rung 5 stack, with high-speed serial links between packages" — the trade press becomes readable. That's the whole goal of this document.

9. Closing thought

SemiAnalysis's instinct in their fourth article was right: the categories are dissolving. What they didn't say is that the *reason* they're dissolving is that the category names were always surface marketing. The underlying physics has one axis — *interconnect density*, measured in pitch — and one secondary axis — *dimensionality*. Every product is a point in that 2D space, and every vendor sells multiple products across multiple quadrants of it.

If you remember one thing from this document, make it the ladder. If you remember two things, add the bonding-method cross-product. If you remember three, add the substrate-material choice (organic, silicon, glass) that sits inside each rung. Everything else — brand names, "2.1D vs 2.3D," which vendor's RDL-first process yields 3% better — is detail on that frame.

The next decade of this industry will be about pushing every rung's cost curve down further, so multi-rung packages become standard in lower-margin products. Expect hybrid bonding to appear in laptops. Expect silicon bridges in consumer GPUs. Expect the vocabulary to keep multiplying. But the ladder won't change; only the price tags on each rung will.